

Leveraging Geographic Information and Social Indicators for Misbehavior Detection in VANETs

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Abstract—With the development of Vehicular Ad hoc Networks (VANETs), the security threats to the communication of VANETs are also increasing. Misbehaviors including fake messages attacks, characterized by messages containing misleading traffic information will significantly interfere with the cooperation between vehicles, and reduce the message transmission and service capabilities of VANETs. Therefore, it is necessary to propose an effective misbehavior detection scheme to ensure the reliable communication of VANETs. In addition to considering the dynamic network topology of VANETs, the social relationship between vehicles can also be applied as a relatively stable indicator to evaluate vehicles. This paper proposes a misbehavior detection scheme based on geographic information in vehicular social networks, to address potential security threats in VANETs. The proposed scheme constructs social relationships by analyzing similarities in geographic movement patterns of vehicles, and also evaluates the message trust from time and spatial factors. Finally, the proposed scheme integrates message trust with the established social relationships, enhancing the overall misbehavior detection accuracy, and then utilizes a reputation system to dynamically adjust the involvement of vehicles in misbehavior detection based on their historical behavior. Moreover, simulation results extensively demonstrate that our proposed scheme effectively detects malicious vehicles, performing consistently well across different ratios of such vehicles and movement patterns.

Index Terms—Detection Scheme, Geographic Information, Message Trust, Social Relationship.

I. INTRODUCTION

The emergence of Vehicular Ad hoc Networks (VANETs) has brought great convenience to transportation, which can significantly improve road safety and traffic efficiency, partially support autonomous driving, and provide infotainment services [1]. Smart vehicles within VANETs are equipped to connect with consumer electronics, effectively integrating both information and entertainment systems into the driving experience [2]. These applications in VANETs depend on message exchanges among participants, encompassing both traffic control-related safety aspects and entertainment-oriented non-safety applications [3]. However, once messages are compromised, applications that rely on messages will make wrong decisions. These may result in serious repercussions, threatening the safety of both drivers and pedestrians (e.g., interference

with electronic control units causing brake failure or engine shutdown) [4], [5]. For example, Keen Lab found 14 common security vulnerabilities in the electronic control units of various BMW models, involving in-vehicle infotainment systems, communication modules, and gateways [6]. Therefore, it is urgent to solve the problem of securing the communication of VANETs by adopting information security technologies.

In literature [7]–[12], misbehavior detection scheme that target the identification of attacker misconduct are widely classified into two categories: node-centric and data-centric. The node-centric schemes evaluate the trustworthiness of nodes, considering factors such as the interactions among network participants and the accuracy of historical messages [7]–[9]. For example, speed factor [7] and data forwarding rate [8] are adopted to verify the trust of nodes. The data-centric schemes focus on assessing the authenticity of data itself, independent of its source. This approach typically involves evaluations based on the plausibility and consistency of the data [10]–[12]. For instance, literature [12] verifies message consistency by contrasting the traffic flow reported by a single node against the flow observed by neighboring nodes.

Although significant progress has been made in both node-centric and data-centric schemes, the aspect of social relationships among nodes is often overlooked in misbehavior detection. These relationships, stemming from node interactions and resilient to the dynamic network topology, serve as a potential metric for evaluating node behavior. While the majority of detection schemes prioritize evaluating node trust and data integrity based on the communication layer, including forwarding behaviors and data validity, the incorporation of nodal geographic information into these assessments is often neglected. Furthermore, existing detection schemes typically assume that nodes move randomly, whereas node movements are actually purposeful, such as moving to a shopping mall or company. Therefore, the geographical information of nodes implies the interest preferences of driver, and also effectively reveals their relation to specific events, since the trajectory of node can indicate whether it has passed the event location, or how close it is associated to the event. Social relationships between vehicles can be well constructed based on similar geographical information, thereby maintaining social trust between vehicles.

Our previous work in [13] has investigated social relationships in a single traffic event scenario. In that work, trajectory similarity, contact frequency, and message trust were studied based on geographic information. Results have demonstrated the effective detection rate of previous work at various mali-

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cious vehicle proportions and mobility patterns. In addition, our previous work and other works focus on detecting a single complete event, while lacking consideration of complex traffic scenarios (e.g., multiple events with different impacts on urban traffic). Therefore, this paper further investigates the influence of geographic information on social relationships in complex traffic scenarios (i.e., the multi-event scenario over long-term domains). The major contributions of this paper are summarized as follows.

- 1) This paper proposes a geographic information-based misbehavior detection scheme for the multi-event scenario in the long-term domain. To cope with the actual complex traffic, this paper combines multi-events with the reputation scheme to adjust social relationships, thereby dynamically adjusting the involvement of vehicles in misbehavior detection. In addition, this paper introduces negative evaluations in calculating social relationships combined with vehicular reputation.
- 2) Considering the time attribute of geographic information, this paper introduces geographical region popularity, trajectory time sequences similarity, and historical geographic preference to measure the similarity between vehicles. The trajectory time sequences similarity applies the dynamic time warping technology, which calculates the similarity, by extending or shortening the time sequence to corresponding points of two sequences.

The rest of this paper is structured as follows. Section II reviews the related work. Section III describes the system model. Section IV presents our proposed scheme. Section V reports evaluation results. Section VI concludes this paper.

II. RELATED WORK

A. Node-Centric Scheme

The node-centric scheme verifies node behavior by analyzing the frequency of data exchange and forwarding behavior of network participants.

1) *Behavior-based Scheme*: The behavior-based scheme mainly focuses on observable nodal behavior and extracts metrics to evaluate the nodal trust. A typical work is the watchdog scheme [14]. The watchdog scheme involves each node monitoring its neighbor nodes to ensure that they actually forward the data that should be forwarded. Jedari et al. [8] propose a watchdog scheme based on social relationships. Based on the social relationship with encountered nodes, the watchdog scheme infers their message forwarding behavior and identifies their types (i.e., cooperative, individually selfish, and socially selfish).

2) *Trust-based Scheme*: Nodes with a history of good reputations are often more reliable in the future, such schemes are effective at distinguishing between honest and malicious nodes. Li et al. [9] propose an attack-resistant trust management scheme. Their work collects data from multiple nodes and applies a collaborative filtering algorithm to calculate the reliability of nodes, based on functional trust and recommended trust factors. However, the accuracy of their work degrades when the network lacks sufficient interaction. Tan et al. [15] apply fuzzy logic and graph theory algorithms to

evaluate the routing reliability of nodes. Their work considers two trust factors, i.e., packet transmission rate and average delay time.

B. Data-Centric Scheme

The data-centric scheme focuses on the content of data to determine its validity, including plausibility checks and consistency checks.

1) *Plausibility Checks*: Jaeger et al. [11] utilize the Kalman filter to track the real-time position of nodes by analyzing their historical location, speed, and driving direction as detailed in cooperative awareness messages. Their work predicts the position range of node in the following time period, thereby checking the plausibility of node location contained within cooperative awareness messages. Baza et al. [16] detect Sybil attacks based on proofs of work and location verification. Their work considers the trajectory of node as its unique identity, and receives location proofs from multiple roadside units by way of threshold signature to generate a unique trajectory. To restrict malicious nodes from generating multiple trajectories at the same time (i.e., restricting having multiple identities), nodes must complete proofs of work before receiving location proofs from location proofs. Meanwhile, for malicious nodes with high computing power (e.g., cloud services), their work applies the maximum clique technique to identify multiple similar trajectories created by malicious nodes.

2) *Consistency Checks*: Zaidi et al. [12] comparative analysis between the traffic flow observed from a single node and that from its neighboring nodes. Their work considers that the traffic flow from nodes close to each other under the same traffic conditions should be very similar. If there is an attacker faking the content of message, there will be a significant difference in the traffic flow from the normal node and the malicious node. Antonopoulos et al. [17] propose a selfish behavior detection scheme based on two statistical methods, including the Kruskal-Wallis test and Conover-Iman multiple comparisons. The Kruskal-Wallis test analyzes whether there are at least two types of behaviors (i.e., normal behavior and misbehavior) in the network, based on the inconsistency of forwarding behavior between normal and malicious nodes. The Conover-Iman method performs multiple comparisons for inconsistent behaviors and finally identifies malicious nodes.

C. Vehicular Social Networks

Connected vehicle communication can be seen as “the first social network for vehicles”, since each driver can share data with neighbors. For example, certain areas on the road (e.g., intersections, business districts, etc.) often experience traffic congestion, which can be shared through VANETs. As a result, roads become popular social places where vehicles communicate with each other [18]. Since VANETs mainly serve applications of entertainment information and traffic safety, human factors are involved. Therefore, social characteristics and human behaviors largely affect the performance VANETs, resulting in the vehicular social networks.

The vehicular social networks consist of a group of vehicles sharing common interests or preferences. Meanwhile,

they are on the road close in time and space dimensions. Specifically, in addition to the communication protocol, the vehicular social networks include human factors that affect vehicle connectivity, namely human mobility, selfishness, and user preferences. A social-based protocol can identify nodes that are socially similar or share common interests, e.g. a group of people driving to a football match can meet on the same traffic scene, and also have a strong desire to meet other people with similar interests. In general, vehicles can publish and share a lot of valuable information with others, such as personal information (i.e. location, destination, voice notes, pictures, etc.), and traffic information (i.e. accidents, road construction, congestion, etc.). For example, Sha et al. [19] design NaviTweet publishing or listening to traffic-related voice tweets, to customize driving routes for drivers.

There are also numerous works on vehicular social networks to facilitate message forwarding. Xia et al. [20] propose an interest-based forwarding routing protocol inspired by artificial bee colonies. Their work divides nodes into different communities based on node interests, and records the information (e.g., node density) of nodes passing through communities. The density information provides guidance for selection of next-hop forwarding nodes. Gong et al. [21] propose a social contribution-based routing protocol, exploiting (i) the probability of delivering a message to a target node based on the social relationship between vehicles, and (ii) the social contribution of relay nodes. Among them, the social contribution is applied as a key factor to stimulate selfish vehicles to cooperate well in the network. Based on these two metrics, vehicles with higher delivery probability and lower social contribution are selected as next-hop forwarders.

III. SYSTEM MODEL

A. Network Model

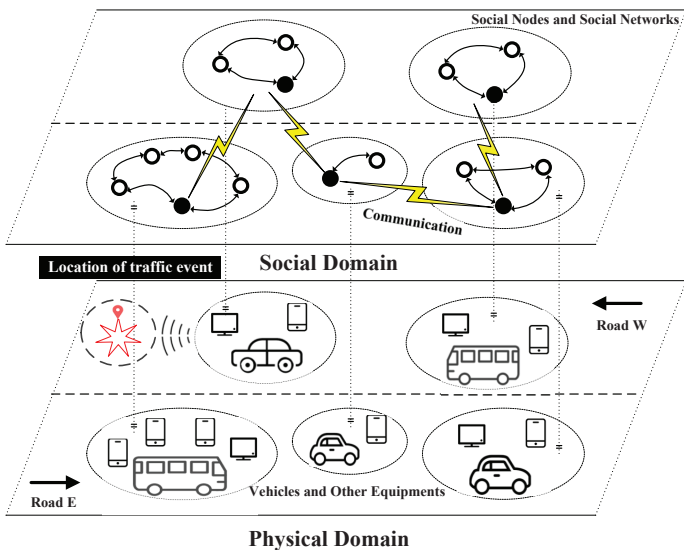


Fig. 1: Overlapping physical and social domains. Vehicles (filled circles) construct social relationships based on mobility and shared interests.

TABLE I: Traffic event levels and durations

Levels	Durations
I	12 min
II	18 min
III	24 min
IV	30 min
V	36 min

Fig. 1 represents two layers of mutual mapping in the vehicular social network, namely the physical domain and social domain. The physical domain is modeled as a general VANETs model with N vehicles. Based on existing works, this paper posits the implementation of public key-based authentication systems, encompassing public-private key pairs, digital certificates, and a certificate authority (CA). The CA issues certificates to entities to sign messages proving their identity. Then, this paper evaluates the trust of internal vehicle messages, as described in Section IV-A.

The social domain consists of one-way social relationships, since the social relationship of i -th vehicle to j -th vehicle may be different from that of j -th vehicle to i -th vehicle, due to their different social situations. In addition, different from [8], [22], (based on real-world datasets and physical proximity), this paper measures the social relationship between vehicles according to their geographic information. The detailed calculation method will be introduced in Section IV-B.

Considering the urban road traffic, this paper adopts the event-driven message to notify traffic events, denoted as m . The event-driven message includes the location of vehicle as $\mathcal{L}_i$, the time of event as $\mathcal{E}_t$, and the location of event as $\mathcal{E}_l$. Furthermore, event-driven messages received by the i -th vehicle from encountered vehicles are added to its message set $\mathcal{M}_i$. Considering multiple traffic events in the long time domain, this paper defines traffic events that cause traffic congestion with different severity levels and durations. The traffic event with high severity level will cause longer periods of congestion in larger regions, and its severity level $\mathcal{S}_E$ is defined as shown in Table I. Meanwhile, Section IV-C will punish or restore corresponding vehicles based on traffic event levels and detection results, due to the impact of traffic event levels on driver judgment and traffic conditions.

B. Problem Definition

When a vehicle is notified of a traffic event through event-driven messages, it will respond, for example, by re-routing to avoid the location of traffic event about congestion while driving. However, vehicles may be misguided by fake messages to respond incorrectly, due to the existence of misbehavior. Fig. 2 shows a malicious vehicle reporting a fake message about a traffic event at location X. At this time, vehicles with notified fake event will adjust their driving paths to avoid passing the event location, decreasing overall traffic efficiency. Such a malicious vehicle is a type of internal attacker, who already has passed the authentication but can participate in network communications to misguide other vehicles. Internal attackers have the capability to perform various malicious behavior to interfere with the judgment of normal vehicles. For example, an internal attacker can send fake messages containing

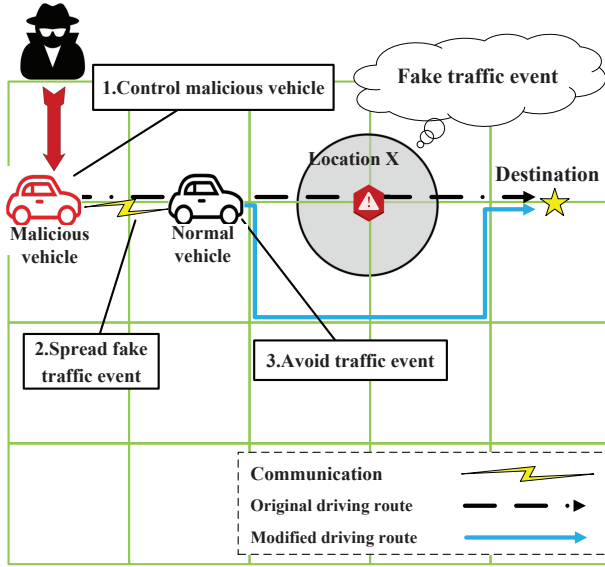


Fig. 2: A misbehavior example of fake traffic event.

misleading traffic information to other vehicles, or choose to discard traffic messages to interfere with the judgment of normal vehicles regarding traffic conditions. Furthermore, internal attackers can also deliberately accuse normal vehicles of being malicious, or alternatively perform normal and malicious behaviors to undermine the trust of normal vehicles in other vehicles, thus affecting the cooperation between vehicles in VANETs. As a result, the overall performance of transportation system will be seriously affected. In addition, this paper assumes a circular message reference range with the event location as the center and Φ as the radius. Messages reported by vehicles far away from the event (i.e., not within the Φ range) will not participate in the detection.

C. Attack Model

The attacker model proposed by Raya et al. [23] is an important research in VANETs. It expounds various attack modes, and classifies attackers according to the following three classification dimensions: a) Based on whether the attacker has valid credentials, attackers can be classified as internal and external attackers. b) Based on whether the attacker can directly benefit from the attack, the motivation of attacker can be classified as rational and malicious. c) Based on whether the attacker sends a message, the attack behavior can also be classified as active and passive.

This paper primarily addresses internal attackers who possess legitimate certificates issued by the CA. Combining the characteristics of VANETs and attacker models, misbehaviors launched by internal attackers are represented as follows.

- **Fake message attack:** Attackers actively forge traffic events that did not actually occur for selfish purposes, aiming to spread misleading traffic information within the network and facilitate their own travel.
- **Gray hole attack:** Attackers selectively and passively drop messages to interfere with the decision-making of

normal vehicles. For example, attackers may strategically drop messages from normal vehicles to impede message forwarding in the network.

- **Bad-mouth attack:** Attackers actively forge and share social relationships with encountered vehicles to destroy the reputation of other vehicles, aiming to maliciously manipulate and destroy the reputation of other vehicles.
- **Time-shift attack:** Attackers alternately perform normal and malicious to interfere with the detection, aiming to make itself detected as normal by other vehicles.

D. Scheme Overview

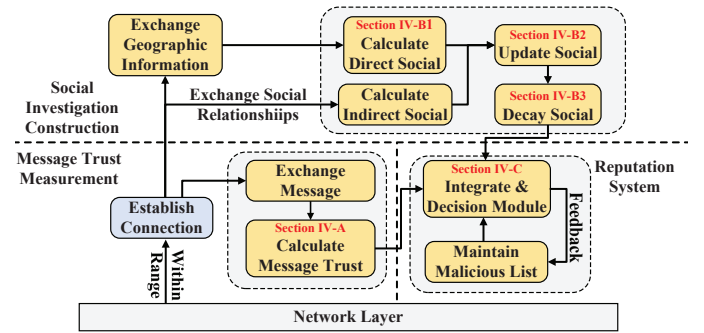


Fig. 3: The architecture of proposed scheme.

In Fig. 3, the architecture of this paper consists of three components: message trust measurement, social investigation construction, and reputation system. The message trust measurement involves the vehicle (e.g., i -th vehicle) calculating the trust of received messages, by considering the spatial and time of message content. In the social investigation construction, this paper considers the regional popularity, the trajectory time sequences similarity, and historical geographic preference to calculate the direct social relationship. Meanwhile, considering the transitivity of social relationship, the indirect social relationship is considered. Then, the reputation system combines message trust and social relationships to inform decisions regarding traffic events. In addition, it updates the reputation to dynamically adjust the vehicular reputation, thereby adjusting the ability of vehicles detected as malicious to participate in VANETs communications. Based on the historical reputation, the reputation system introduces negative evaluation into the indirect social relationship. Table II summarizes the widely used notations.

IV. DESIGN OF PROPOSED SCHEME

A. Message Trust Measurement

The message trust calculation considers the spatial and time factors of message content.

a) Spatial Factor

In reality, when a traffic event occurs, vehicles close to the event location issue messages to inform other vehicles to adjust their driving paths. The existing works [24] have demonstrated that the distance in meters between the vehicle location $\mathcal{L}_i$ and the event location $\mathcal{E}_l$ in m , is a crucial factor in assessing the

TABLE II: Notations

Notations	Descriptions
m	The message driven by a traffic event.
$\mathcal{E}_t$	The time when the vehicle creates a message m .
$\mathcal{E}_l$	The event location contained in m reported by vehicles.
$\mathcal{L}_l$	The vehicle location contained in m when the vehicle reports an event.
$\mathcal{M}_i$	The set of event messages received by i -th vehicle.
$\mathcal{S}_E$	The severity level of traffic event.
$\mathcal{R}_a$	The number of vehicles passing through the region a .
$\mathcal{C}_{i,a}$	The number of visits of i -th vehicle in region a .
$\mathcal{N}_i$	The set of vehicles having a social relationship with i -th vehicle.
MT_m	Message trust value of message m .
P_a	The regional popularity of region a .
$ED_{ij}(x, y)$	The Euclidean distance between the x -th location of i -th vehicle trajectory and the y -th location of j -th vehicle trajectory
$TS_{ij}(x, y)$	The trajectory time sequences similarity between the first x locations of i -th vehicle and the first y locations of j -th vehicle.
GP_{ij}	The similarity between the i -th vehicle and j -th vehicle in historical geographical preferences.
DS_{ij}	The direct social relationship between i -th vehicle and j -th vehicle.
I_{ij}	The importance of i -th vehicle assigned to j -th vehicle.
IS_{ij}	The indirect social relationship between i -th vehicle and j -th vehicle.
SR_{ij}	The social relationship between i -th vehicle and j -th vehicle.
DV_i	The detection value of i -th vehicle.
MV_{ij}	The malicious value of j -th vehicle with respect to i -th vehicle.

trust of message. Meanwhile, the farther the distance between the vehicle and event, the lower the trust of message.

However, these works did not discuss the calculation method of distance in detail. In fact, due to the influence of obstacles (e.g., high-rise buildings and green trees) in urban traffic, the Euclidean distance cannot reflect the actual distance of vehicle when driving. The Manhattan distance is often applied to calculate the distance between two points on urban roads. It represents the sum of absolute values of coordinate differences between two points on the standard coordinate system, that is, the actual driving distance of vehicle from one point to another [24]. The Manhattan distance has high applicability when calculating message trust in this paper. Therefore, this paper considers the Manhattan distance ϵ between the vehicle location $\mathcal{L}_l$ and the event location $\mathcal{E}_l$ as the spatial factor for calculating the message trust.

b) Time Factor

Due to the non-real-time characteristics of communication in VANETs, the time when the vehicle detects received the message may have a significant difference from the event time reported by the message. Meanwhile, traffic events are ultimately addressed by the traffic department. The probability of traffic events being dealt increases with time elapsed. Therefore, this paper measures the message trust by comparing the event occurrence time with the detection time of vehicle. Since most vehicles cannot obtain the exact time of traffic event (except for a few vehicles that observe the event), this paper selects the earliest time from these messages to estimate the event occurrence time δ_t . Subsequently, the discrepancy between the event time $\mathcal{E}_t$ noted in the message and the estimated event time δ_t , is then utilized as a time factor in the calculation of message trust.

Therefore, the trust value of message m indicated by lines 3-10 of Algorithm 1, denoted as MT_m , is calculated as follows:

$$MT_m = \beta_1 * e^{-\alpha_1(\mathcal{E}_t - \delta_t)} + (1 - \beta_1)e^{-\alpha_2 * \epsilon / \Phi} \quad (1)$$

where $\mathcal{E}_t$ and δ_t are the event time in the message and approxi-

mate event time respectively, ϵ is the Manhattan distance from $\mathcal{L}_l$ to $\mathcal{E}_l$, α_1 and α_2 set the change of exponential function, β_1 controls the influence of time and spatial factors, and Φ is the message reference range, which balances the excessive Manhattan distance. Algorithm 1 represents the pseudo-code for calculating the message trust of i -th vehicle.

Algorithm 1: Message Trust Measurement

Input : Message set of i -th and j -th vehicles as $\mathcal{M}_i$ and $\mathcal{M}_j$, Event time $\mathcal{E}_t$, Event location $\mathcal{E}_l$, Vehicle Location $\mathcal{L}_l$

Output: Message trust set $\{MT_m | m \in \mathcal{M}_i\}$

- 1 Upon i -th vehicle establishes a connection with j -th vehicle;
- 2 Receive message set $\mathcal{M}_j$ from j -th vehicle;
- 3 $\delta_t \leftarrow \text{Max}$;
// Initialize event time
- 4 **for** Message $m \in \text{Message set } \mathcal{M}_i$ **do**
- 5 $\delta_t \leftarrow \text{Min}(\delta_t, \mathcal{E}_t \text{ in } m)$;
 // $\mathcal{E}_t$ is the time when the vehicle creates the event message m
- 6 **end**
- 7 **for** Message $m \in \text{Message set } \mathcal{M}_i$ **do**
- 8 $\epsilon \leftarrow \text{Calculating Manhattan distance of } \mathcal{L}_l \text{ and } \mathcal{E}_l$;
- 9 $MT_m \leftarrow \text{Calculating the message trust based on time and spatial factors}$;
- 10 **end**
- 11 **Return** $\{MT_m | m \in \mathcal{M}_i\}$;

B. Social Investigation Construction

In VANETs, the vehicle is usually operated by drivers, and typically has frequent driving destinations or fixed driving routes (e.g., a return route from home to work). Analyzing the geographic information generated by vehicle movement can help to explore the geographic similarity between vehicles, thereby constructing the social relationship between vehicles. Therefore, this paper takes geographic information as an important factor in constructing direct social relationships. Meanwhile, based on the transitivity of social relationships, this paper introduces indirect social relationships to explore hidden attributes in VANETs, thus establishing better social relationships between vehicles.

1) Direct Social Relationship

Our previous work considered geographic information, by simple calculating the trajectory similarity between vehicles through cosine similarity [13]. Compared with the previous work, the model in calculation of direct social relationship has been improved in this paper. It applies an improved geographic information calculation method to explore the potential similarity between vehicles. Specifically, this module constructs direct social relationship based on the geographical regional popularity, trajectory time sequences similarity, and historical geographical preference.

a) Geographical Region Popularity

In order to reflect the characteristics of geographical region, this paper constructs a model that encodes the regional charac-

teristics into the process of calculating geographic information. Based on Tobler's first law [26] "everything is related to everything else, but near things are more related than distant things", vehicles with similar characteristics are likely to cluster into the same geographic region. To simulate the influence of geographic features on vehicle aggregation, this paper divides the map into multiple virtual regions (represented in Section V-A), and introduces the concept of geographical region popularity [27].

Definition 1. Geographical Region Popularity: The number of vehicles passing through the region per unit time.

Usually, regions containing popular shopping malls or tourist attractions attract mass vehicle visits, thus having high geographical region popularity. However, the behavior of visiting high popularity regions is difficult to reflect the characteristics of vehicles, since mass vehicles visit high popularity regions due to personal needs. Therefore, this paper considers that the behavior of two vehicles jointly visiting a low popularity region can better reflect the similar social characteristic of two vehicles.

In addition, when considering geographic factors later, the reciprocal of geographical region popularity will be applied as the weight to encode the calculation process of geographic information. Due to distributed nature of VANETs, it is difficult for vehicles to obtain the global regional popularity from nodes similar to the center. Therefore, the geographical region popularity is calculated by analyzing the trajectory information shared by encountered vehicles. Furthermore, this paper defines a time window to measure the geographical region popularity in the recent period. The popularity P_a of region a is the number of vehicles passing through the region a per unit time in the past period of time window Ω , calculated as follows:

$$P_a = \frac{\mathcal{R}_a}{\Omega} \quad (2)$$

where Ω represents the size of time window, and $\mathcal{R}_a$ represents the number of vehicles passing through the region a within the time window.

b) Trajectory Time Sequences Similarity

Trajectory sequences containing time attributes are high-dimensional data. Previous works often applied the Euclidean distance to measure the similarity between two trajectory sequences. However, since the trajectory sequence is affected by the time dimension, the compared two trajectory sequences may have inconsistent lengths, or only have offset differences on the time axis (i.e., they can be overlapped by translation). The Euclidean distance only considers the distance between two locations on the sequence, without considering the duration of each location. Thus, the Euclidean distance cannot effectively measure the similarity between the two trajectory time sequences. Dong et al. [28] summarized inadequacies of Euclidean distance, including (i) it cannot discern the similarity of sequences shape, (ii) it cannot reflect the similarity in magnitude of trend dynamic change between two sequences, (iii) the calculation based on point distance cannot reflect the difference between different analyzed frequencies.

To address the offset of same trajectory in the time domain, this paper adopts Dynamic Time Warping (DTW) [29] technology to measure the similarity between two trajectory time sequences. The DTW calculates the similarity between two trajectory sequences by extending or shortening sequences, and matching the location with multiple consecutive locations in another sequence. Therefore, the DTW can calculate the similarity of sequences with different lengths, and suitable for sequences with time offsets.

Fig. 4 represents the process of DTW matching two time sequences, where dotted lines represent the corresponding similarity locations between two trajectory sequences. Fig. 4(a) represents the original two trajectory sequences, while Fig. 4(b) extends similar locations in the two sequences. For example, the location A of j -th vehicular trajectory is extended in time to correspond to two consecutive locations A of i -th vehicular trajectory. The DTW measures the similarity based on the sum of distances between all these corresponding locations, called the normalized path distance. In the actual solution, the DTW obtains the minimum distance between two sequences, by finding the minimum alignment matching path. Meanwhile, the minimum alignment matching path is based on the distance matrix of two sequences and the dynamic programming method. Therefore, the trajectory time sequences similarity between i -th and j -th vehicles is expressed as $TS_{ij}(x, y)$. The recursive formula for calculation is expressed as follows:

$$TS_{ij}(x, y) = ED_{ij}(x, y) + \min(TS_{ij}(x-1, y), TS_{ij}(x, y-1), TS_{ij}(x-1, y-1)) \quad (3)$$

where $ED_{ij}(x, y)$ represents the Euclidean distance between the x -th location of i -th vehicle trajectory sequence and the y -th location of j -th vehicle trajectory sequence, $TS_{ij}(x, y)$ represents the similarity between the first x locations of i -th vehicle and the first y locations of j -th vehicle.

c) Historical Geographic Preference

The vehicular trajectory usually implies the vehicular preferences for different geographic regions, which contributes to constructing social relationships between vehicles. However, it is difficult to directly obtain the vehicular preference for geographic regions. Therefore, the vehicular implicit social preference can be explored by analyzing historical visit behaviors of vehicles [30]. This paper denotes the number of times that i -th vehicle visits the region a as $\mathcal{C}_{i,a}$. When measuring the similarity of vehicles in historical geographic preferences, several similarity measures can be applied, such as cosine similarity and Pearson correlation. For convenience, this paper chooses the cosine similarity to calculate the similarity between i -th and j -th vehicles in historical geographical preferences. Meanwhile, the reciprocal of regional popularity is considered as the regional weight into the calculation. The calculation formula is as follows:

$$GP_{ij} = \frac{\sum_{a \in V} (\mathcal{C}_{i,a} \cdot \mathcal{C}_{j,a} / P_a)}{\sqrt{\sum_{a \in V} \mathcal{C}_{i,a}^2} \cdot \sqrt{\sum_{a \in V} \mathcal{C}_{j,a}^2}} \quad (4)$$

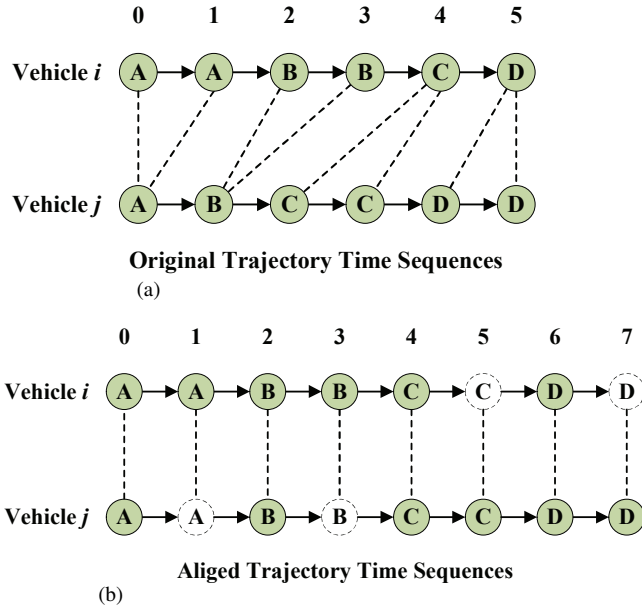


Fig. 4: (a) The original two trajectory time sequence should be aligned. (b) Already aligned two trajectory time sequence.

where $C_{i,a}$ and $C_{j,a}$ represent the number of times that i -th vehicle and j -th vehicle visit the region a , respectively. V represents the total number of divided virtual regions.

Therefore, the direct social relationship between j -th vehicle and i -th vehicle is calculated as follows:

$$DS_{ij} = TS_{ij}(x, y) + GP_{ij} \quad (5)$$

In the follow-up simulation experiments, the sum of two features is applied as the benchmark for the direct social relationship. Under realistic conditions, weight values can be assigned differently based on the significance of trajectory time sequences similarity and historical geographic preference.

2) Indirect Social Relationship

The direct social relationship between the i -th and j -th vehicles relies on limited direct encounters (limited chances of encountering while driving). Therefore, it cannot fully reflect the reputation of j -th vehicle in the social network. For example, j -th vehicle deliberately encounters i -th vehicle to maintain the direct social relationship between them, but i -th vehicle cannot obtain the reputation of j -th vehicle in other vehicles. Hence, this paper investigates the indirect social relationship between the i -th and j -th vehicles through their mutual friends (e.g., v -th vehicle).

In social networks, metrics like degree centrality and eigenvector centrality play key roles in assessing the vehicular significance. Degree centrality measures the vehicular connectedness through the number of its connections with others. Eigenvector centrality evaluates the vehicular importance, considering not only the number of connections but also the relevance of those connected vehicles. Since the vehicular social network is also a kind of social network, it has the characteristics of degree centrality and eigenvector centrality.

This paper defines the degree centrality of vehicle as the initial importance of vehicle. Meanwhile, based on the char-

acteristic of eigenvector centrality, this paper assigns the initial importance of vehicle to other vehicles. Inspired by the PageRank algorithm [31], this paper designs a calculation method for indirect social relationships. Specifically, the vehicle assigns the initial importance to the evaluated vehicle, according to the relative social relationship of evaluated vehicle among all social relationships. In general, if a vehicle establishes social relationships with many other vehicles, the importance of this vehicle is considered to be relatively high. If a vehicle with high importance establishes social relationship with other vehicles, the importance of these vehicle will also increase accordingly. Based on the social relationship between the v -th and j -th vehicle (represented as SR_{vj} and initially assigned as 0), the significance of v -th vehicle is attributed to j -th vehicle when $SR_{vj} > 0$, calculated as follows:

$$I_{vj} = \frac{SR_{vj}}{\sum_{h \in \mathcal{N}_v} SR_{vh}} |\mathcal{N}_v| \quad (6)$$

where $|\mathcal{N}_v|$ is the size of social vehicles set $\mathcal{N}_v$, i.e., the importance of v -th vehicle.

In addition to the importance I_{vj} assigned by vehicle v -th to j -th vehicle, this paper also considers the social relationship between friend v -th and i -th vehicles as the correction factor. If the social relationship between the i -th and v -th vehicles is weak, the v -th vehicle's influence on the relationship between the i -th and j -th vehicles is limited, even if v -th vehicle has a strong importance. Consequently, the indirect social relationship between the j -th and i -th vehicles, is defined by considering the importance assigned by all their mutual friend vehicles, as follows:

$$IS_{ij} = \sum_{v \in \mathcal{N}_i} (SR_{iv} * I_{vj}) \quad (7)$$

Finally, the social relationship between the i -th and j -th vehicles is calculated as follows:

$$SR_{ij} = DS_{ij} + IS_{ij} \quad (8)$$

where the sum of DS_{ij} and IS_{ij} is applied as the benchmark for the social relationship. Under realistic conditions, weight values can be assigned differently based on their significance.

3) Social Decay

The maintenance of social relationships between vehicles relies on regular encounters, and is thus subject to temporal factors. When any pair of vehicles have not encountered for a long time, their social relationship will decay over time, since vehicles trust the recently encountered vehicle more. Therefore, this paper proposes a social relationship decay model as follows:

$$SR_{ij} = SR_{ij} * \Gamma^{(\zeta_t - \eta_t)/\Delta} \quad (9)$$

where $\Gamma < 1$ sets the size of each decay, Δ sets how long to decay, η_t is the last updated time of SR_{ij} , and ζ_t is the current time. Algorithm 2 presents the pseudo-code for constructing social relationships on the i -th vehicle.

Algorithm 2: Social Investigation Construction

Input : The number of vehicles passing through the region a $\mathcal{R}_a$, The number of visits of i -th vehicle in region a $\mathcal{C}_{i,a}$, The Euclidean distance between the x -th location of i -th vehicle trajectory and the y -th location of j -th vehicle trajectory $ED_{ij}(x, y)$

Output: Social relationship as SR_{ij}

- 1 Upon i -th vehicle establishes a connection with j -th vehicle;
- 2 $P_a \leftarrow$ Calculating the geographic region popularity by the number of visits;
- 3 $TS_{ij}(x, j) \leftarrow$ Calculating the trajectory time sequences similarity according to vehicle trajectories;
- 4 $GP_{ij} \leftarrow$ Calculating the historical geographic preference according to $\mathcal{C}_{i,a}$, $\mathcal{C}_{j,a}$, and P_a ;
- 5 $DS_{ij} \leftarrow$ Calculating the direct social relationship according to Eq. (5);
- 6 $IS_{ij} \leftarrow$ Calculating the indirect social relationship according to Eq. (6) and Eq. (7);
- 7 $SR_{ij} \leftarrow$ Calculating the social relationship according to Eq. (8) and the decay model;
- 8 $\eta_t \leftarrow$ Current time;
// Update the social relationship change time
- 9 Return SR_{ij} ;

C. Reputation System

The reputation system mainly includes the integration and feedback scheme. The former mainly integrates the message trust and social relationships to obtain the detection decision, based on the vehicle reputation. The latter mainly dynamically updates and manages the vehicle reputation based on the detection decision.

1) Integration Scheme

Based on Section IV-A and Section IV-B, this paper links message trust and the social relationship of message creators as linked data. Then, the linked data is integrated to make the detection decision. In addition, before integrating message trust and social relationships, this paper divides the received event messages based on different events.

This paper only considers the binary case of whether a traffic event occurs or not. Therefore, this paper posits that the value of message reporting an event is set to 1; otherwise, if it does not report an event, the value is set to -1 . The weighted voting [32] method is applied for integration, and the linked data is applied as the weight factor of message. Therefore, the detection value DV_i obtained by integrating all linked data based on the view of i -th vehicle is as follows:

$$DV_i = \sum_{m \in \mathcal{M}_i} \sigma_m * (\beta_2 * MT_m + (1 - \beta_2) * SR_{im}) \quad (10)$$

where $\sigma_m \in \{-1, 1\}$ sets the value of message m , and β_2 sets the influence of two factors.

When $DV_i > 0$, the i -th vehicle considers that an event

actually occurred. The message m reporting no event, denoted as $\sigma_m < 0$, is inconsistent with the detection value DV_i . The creator of message m is considered to have performed misbehavior (i.e., reporting a fake traffic event). The same applies when $DV_i < 0$. In addition, it is assumed in this paper that vehicles will aggregate messages over each time interval Θ to gather sufficient data for accurate event detection. Algorithm 3 presents the pseudo-code of proposed scheme at i -th vehicle.

Algorithm 3: Proposed Scheme Algorithm

- input :** Message set of i -th vehicle as $\mathcal{M}_i$, Social relationships of i -th vehicle as SR_i , Detection interval Θ , Last detection time τ_t
- output:** Detection Value DV_i
- 1 **if** current time $-\tau_t \geq \Theta$ **then**
 - // τ_t is the last detection time
 - 2 $DV_i \leftarrow 0$;
 - 3 **for** $s \in SR_i$ **do**
 - 4 $SR_{is} \leftarrow$ Update the social relationship according to the decay model;
 - 5 **end**
 - 6 **for** Message $m \in$ Message set $\mathcal{M}_i$ **do**
 - 7 $MT_m \leftarrow$ Calculating the message trust according to Eq. (1);
 - 8 $DV_i += \sigma_m * (\beta_2 * MT_m + (1 - \beta_2) * SR_{ij})$;
// Creator of m as j -th vehicle
 - 9 **end**
 - 10 **for** Message $m \in$ Message set $\mathcal{M}_i$ **do**
 - 11 **if** σ_m is inconsistent with DV_i **then**
 - // Creator of m as j -th vehicle
 - 12 $\mid$ Detect j -th vehicle as malicious;
 - 13 **end**
 - 14 **end**
 - 15 $\tau_t \leftarrow$ current time;
 - 16 Return DV_i ;
 - 17 **end**

2) Feedback Scheme

In order to maintain the stability of VANETs and punish malicious vehicles, this paper introduces a feedback scheme, including negative feedback and positive feedback. The former refers to punishing vehicles upon their misbehavior. The latter refers to the limited recovery of vehicular ability to participate in VANETs, after the punished vehicle performs normal behavior. For each event in a given transport scenario, if i -th vehicle considers that j -th vehicle has performed misbehavior based on the detection value DV_i , then j -th vehicle will be added to the malicious list ML_i and assigned an initial malicious value MV_{ij} . The malicious value of j -th vehicle increases when i -th vehicle is considered to perform misbehavior (increase d each time, denoted as negative feedback factor). Similarly, MV_{ij} decreases when j -th vehicle is considered to perform normal behavior, until it is removed from the malicious list (decrease p each time, denoted as positive feedback factor).

For j -th vehicle in the malicious list, i -th vehicle reduces the linked data of j -th vehicle based on MV_{ij} to alleviate

the influence of j -th vehicle, when integrating message trust and social relationship. Therefore, considering that malicious vehicles may perform normal behaviors or be falsely detected, this paper does not directly exclude malicious vehicles from participating in network communication (message receiving and forwarding). Instead, by establishing a reputation system, this paper limits the influence of malicious vehicles and dynamically adjusts their communication ability.

Since the malicious list records the historical misbehavior, this paper introduces negative evaluations in the indirect social relationships by sharing the malicious list. When j -th vehicle is in the malicious list of friend v -th vehicle, the importance I_{vj} assigned to j -th vehicle by friend v -th vehicle will be set negative to weaken the influence of j -th vehicle. Therefore, the indirect social relationship is calculated as follows:

$$IS_{ij} = \sum_{v \in N_i}^{N_i} (SR_{iv} \cdot I_{vj} \cdot \eta_{vj}) \quad (11)$$

where $\eta_{vj} \in \{-1, 1\}$ indicates whether j -th vehicle is in the malicious list of v -th vehicle.

This paper sets the positive feedback factor smaller than the negative feedback factor, to increase the cost of vehicles implementing misbehavior (i.e., $p < d$). Meanwhile, considering the impact of different event levels, the event level $\mathcal{S}_E$ will be applied as the weight of feedback factor to properly update the malicious value, as shown below:

$$MV_{ij} = \begin{cases} MV_{ij} - \mathcal{S}_E \cdot p, & \text{if perform normal} \\ MV_{ij} + \mathcal{S}_E \cdot d, & \text{if perform misbehavior} \end{cases} \quad (12)$$

V. PERFORMANCE EVALUATION

A. Simulation Setup

This paper evaluates the proposed scheme on Opportunistic Network Environment (ONE) [33], utilizing realistic road data from a $4500 \times 3400 m^2$ area in Helsinki city. The simulation experiment sets random traffic events at random locations every 100s, 200 participating vehicles, 6 hours of simulation time, and 1000s message integration time interval. It is assumed that the communication range of vehicle is 100 meters, and the routing protocol is the Epidemic protocol. Other simulation parameters are shown in Table III.

The simulation experiment assigns four types of POIs and divides the map into virtual square regions, as shown in Fig. 5. Malicious vehicles are assumed to randomly select the destination on the simulation map, and travel according to the shortest path of Dijkstra algorithm. The normal vehicle either chooses POIs as the destination with a preset probability, or randomly selects the destination, and travels according to the shortest path of Dijkstra algorithm. Moreover, each simulation outcome is derived from the average of 10 runs, each conducted with a different random seed.

To performance comparison, we implement both IWOT-V [7] and the research work of Xia et al. [34] in ONE. IWOT-V identifies normal and malicious vehicles using global trust values derived from local trust ratings and their historical

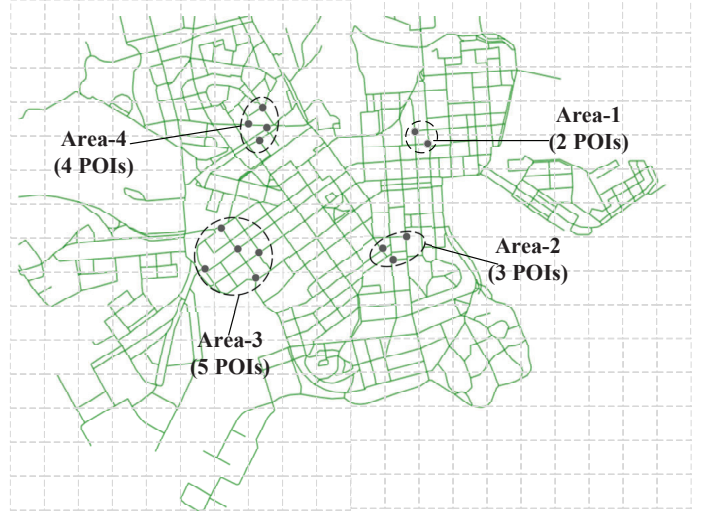


Fig. 5: The Helsinki City Scenario.

interactions. Xia et al. consider two trust attributes of subjective trust and recommendation trust to measure the trust level of evaluated vehicle. The former is measured based on the historical interaction and weighted Markov prediction algorithm. The latter measures the trust of evaluated vehicle based on the feedback scheme and recommender trust. The specific evaluation indicators are described as follows:

Detection Rate: The detection rate refers to the ratio of correctly detected vehicle categories to the total detected vehicle categories, which measures the accuracy of detection scheme, denoted as DR .

$$DR = \frac{TP + TN}{TP + TN + FP + FN} \quad (13)$$

where, TP indicates the number of malicious vehicles correctly identified as malicious vehicles, TN indicates the number of normal vehicles correctly identified as normal vehicles, FP indicates the number of normal vehicles incorrectly identified as malicious vehicles, FN indicates the number of malicious vehicles incorrectly identified as normal vehicles.

Recall Rate: The recall rate refers to the ratio of the number of correctly detected malicious vehicles to the total number of malicious vehicles, denoted as RR .

$$RR = \frac{TP}{TP + FN} \quad (14)$$

Precision Rate: The precision rate refers to the ratio of the number of correctly detected malicious vehicles to the number of considered malicious vehicles, denoted as PR .

$$PR = \frac{TP}{TP + FP} \quad (15)$$

B. Performance Comparison

1) Influence of Malicious Vehicle Ratio

By adjusting the proportion of malicious vehicles, this paper explores the influence of malicious vehicle rate on the detection rate, recall rate, and precision rate. The probability of normal vehicles driving to POIs is fixed at 0.7.

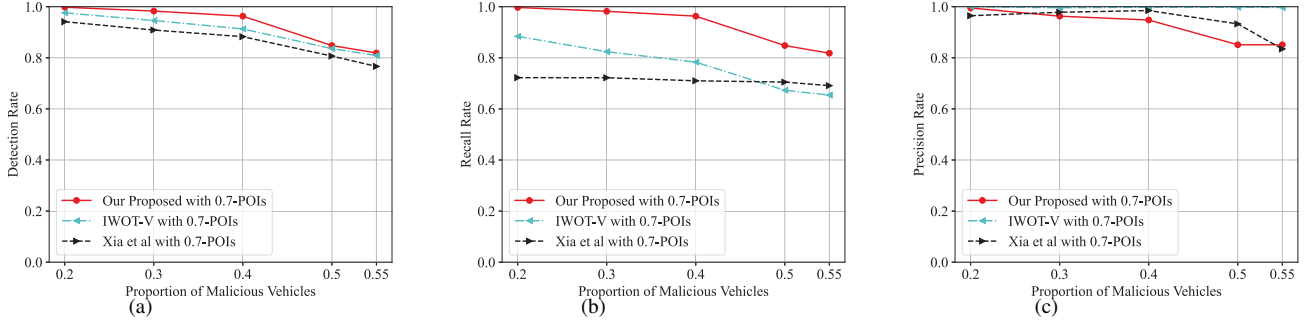


Fig. 6: (a) Detection rate versus the proportion of malicious vehicles. (b) Recall rate versus the proportion of malicious vehicles. (c) Precision rate versus the proportion of malicious vehicles.

TABLE III: Simulation Parameters

Parameters	Parameters value
Simulation area	4500 × 3400m ²
Simulation time	6h
Simulation warm-up	1h
Number of repetition for each run	10
Interval between traffic events	100s
Number of vehicles	200
Transmission range	100m
Waiting time range	[60, 120] s
Speed range	[15, 25] m/s
Routing protocol	Epidemic

Fig. 6(a) shows the influence of malicious vehicle proportion on the detection rate. The detection rate of all schemes decreases as the proportion of malicious vehicles increases. Specifically, when the proportion of malicious vehicles is less than 40%, the detection rates of all schemes are higher than 85%. In the network where the proportion of malicious vehicles is greater than 40%, the messages reported by malicious vehicles occupy a considerable part of network messages, thus the detection rate of all schemes will be significantly reduced.

Fig. 6(b) shows the influence of malicious vehicle proportion on recall rate. The recall rate of all schemes decreases as the proportion of malicious vehicles increases. Specifically, with the proportion of malicious vehicles from 20% to 55%, the recall rate of our proposed scheme decreases from 99.7% to 81.8%, the recall rate of IWOT-V decreases from 88.4% to 65.4%, and the recall rate of scheme in [34] decreases from 72.2% to 71.0%.

Fig. 6(c) shows the influence of malicious vehicle proportion on the precision rate. For our proposed scheme and the scheme in [34], the precision rate of both decreases with the increase of malicious vehicles proportion, while the precision rate of IWOT-V scheme remains stable. Specifically, with the proportion of malicious vehicles from 20% to 55%, the precision rate of our proposed scheme decreases from 99.5% to 85.1%, the precision rate of IWOT-V remains at around 100%, and the precision rate of scheme in [34] decreases from 96.5% to 83.5%. The IWOT-V scheme achieves the best results in precision rate, but is inferior to our proposed scheme in detection rate and recall rate.

The proportion of malicious vehicles has a significant impact on the performance of all schemes. When in network

with few malicious vehicles, all schemes can obtain excellent detection rate, recall rate, and precision rate. However, with the increase of malicious vehicles proportion, the performance of all schemes generally shows a downward trend. Simulation experiments show that similar geographic information helps build better social relationships between vehicles. When normal vehicles have obvious interest in POIs (i.e., POIs probability of 0.7), the performance of our proposed scheme is generally higher than that of the other two schemes. The above simulation results show that the social relationship between vehicles based on geographic information similarity can effectively improve the robustness of system against malicious vehicle attacks.

2) Influence of POIs

By adjusting the probability of normal vehicles driving to POIs, this paper explores the influence of POIs probability on the detection rate, recall rate, and precision rate. The proportion of malicious vehicles is fixed at 50%.

Fig. 7(a) shows the influence of POIs probability on the detection rate. The detection rate of all schemes increases with the increase of POIs probability. Specifically, when the POIs probability increases from 0.3 to 0.7, the detection rate of our proposed scheme increases from 80.4% to 88.0%, the detection rate of IWOT-V increases from 80.9% to 83.6%, and the detection rate of [34] increases from 80.0% to 81.3%.

Fig. 7(b) shows the influence of POIs probability on the recall rate. The recall rate of all schemes increases with the increase of POIs probability, and the change trend is consistent with the influence of POIs probability on the detection rate.

Fig. 7(c) shows the influence of POIs probability on the precision rate. For our proposed scheme and the scheme in [34], their precision increase with the increase of POIs probability, and the change trend is consistent with the influence of POIs probability on the detection rate. The precision rate of IWOT-V scheme is maintained at about 100% during the change of POIs probability, due to its threshold for distinguishing normal vehicles from malicious vehicles. The threshold of IWOT-V may misdetect malicious vehicles as normal vehicles, but with high precision rate of malicious vehicles.

The probability of POIs has a significant impact on the performance of all schemes. It can be seen that the probability of POIs helps to improve the performance of all schemes.

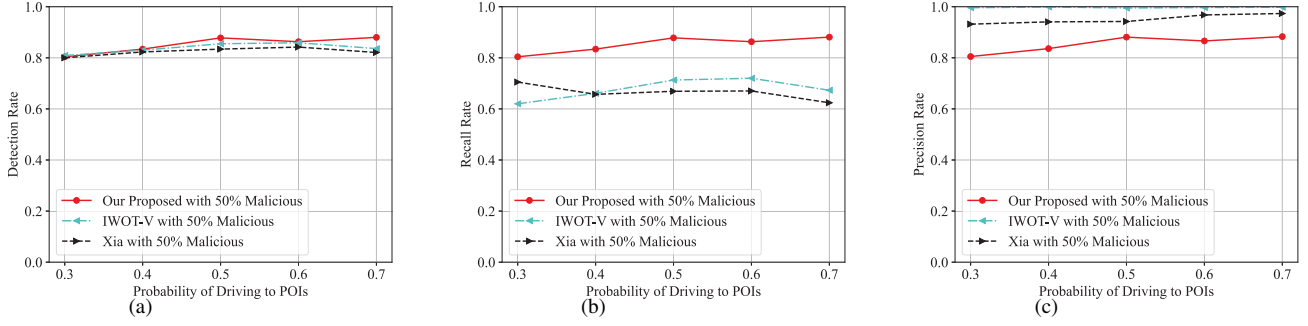


Fig. 7: (a) Detection rate versus the POIs probability. (b) Recall rate versus the POIs probability. (c) Precision rate versus the POIs probability.

Our proposed scheme benefits from a high probability of POIs, enhancing geographical similarity and fostering better social relationships between vehicles. Our proposed scheme is slightly higher than other two schemes in terms of detection rate, much higher than other two schemes in terms of recall rate, and slightly lower than other two scheme in terms of precision rate. However, considering the negative influence of attackers, it is more meaningful to detect actual attackers than to misdetect normal vehicles as attackers. That is, under the condition of high detection rate, high recall rate is more meaningful than high precision rate. Therefore, considering the actual high travel interest and low attack tolerance of driver, our proposed scheme outperforms other two schemes.

3) Influence of Malicious Vehicle Speed

By adjusting the lower bound of malicious vehicle speed, this paper explores the influence of malicious vehicle speed on the detection rate, recall rate, and precision rate. The probability of normal vehicles driving to POIs is fixed at 0.7, the proportion of malicious vehicles is fixed at 50%, the upper bound of malicious vehicle driving speed is fixed at $25m/s$, and the driving speed of malicious vehicles is randomly selected within the experimental range.

Fig. 8(a) shows the influence of malicious vehicle speed on the detection rate. For our proposed scheme and the scheme in [34], their detection rate decrease with the increase of speed lower bound, while the detection rate of IWOT-V scheme increases with the increase of speed lower bound. Specifically, as the lower bound increases from $5m/s$ to $25m/s$, the detection rate of our proposed scheme decreases from 88.2% to 86.8%, the detection rate of scheme in [34] decreases from 83.7% to 81.6%, and the detection rate of IWOT-V increases from 82.9% to 85.9%.

Fig. 8(b) shows the influence of malicious vehicle speed on the recall rate. For the IWOT-V and scheme in [34], their recall rate increases with the increase of speed lower bound, while the recall rate of our proposed scheme decreases with the increase of speed lower bound.

In Fig. 8(c), our proposed scheme and the scheme in [34] both show an increase in precision rate as the speed lower bound increases. However, the precision rate of IWOT-V scheme remains stable.

Therefore, the speed of malicious vehicles has an impact

on the performance of all schemes. The speed of malicious vehicle plays a role in determining the driving range of vehicles and the encounter probability between vehicles, thus affects the calculation of social relationship between vehicles. As for performance indicators, our proposed scheme performs better than other two schemes in terms of detection rate and recall rate, and slightly worse than other two scheme in terms of precision rate. However, as Section V-B2 stated, high recall rate is more meaningful than high precision rate. Therefore, considering low-speed urban traffic, our proposed scheme outperforms other two schemes.

4) Complexity Analysis and Comparison

We analyzed the complexity of these three schemes from four dimensions, as shown in Table IV.

In terms of the architecture, our proposed scheme and Xia et al. [34] are distributed, while IWOT-V adds a central system to the distributed basis. In our proposed scheme and [34], in addition to forwarding traffic messages, only a small amount of control messages needs to be forwarded between vehicles. In IWOT-V, in addition to forwarding messages between vehicles, vehicles also need to wait for messages interaction with the central system, which is more complex than our proposed scheme and [34]. However, these differences are greatly affected by specific traffic scenarios and vehicle encounter opportunities, and are difficult to specifically quantify.

In terms of factors involved, our proposed scheme considers message content and geographical information, while [34] and IWOT-V mainly consider historical interactions.

In terms of the data processing algorithms applied, our proposed scheme applies DTW, [34] applies Weighted Markov Prediction, and IWOT-V applies the Bayesian algorithm.

In terms of time to detect attackers, vehicles in our proposed scheme and [34] directly detect at certain intervals based on existing data, while vehicles in IWOT-V receive detection results from the central system that communicates indirectly.

In addition, we conducted the simulation on a personal computer with 32G memory and i7-7500u CPU. The simulation calculation average time is shown in Table V. During the simulation process, the simulation platform ONE affords the computational overhead of executing the detection scheme every Θ_s (set to 1000s) simulation time interval for each vehicle. Therefore, when these schemes are deployed on a

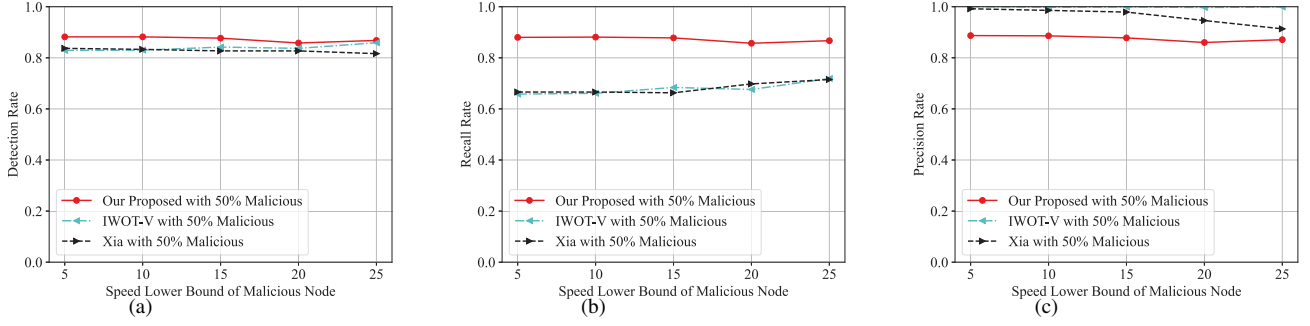


Fig. 8: (a) Detection rate versus the speed of malicious vehicle. (b) Recall rate versus the speed of malicious vehicle. (c) Precision rate versus the speed of malicious vehicle.

TABLE IV: Complexity Analysis and Comparison of Three Schemes

Schemes	Architecture	Factors	Data processing algorithms	Detection time
Our proposed scheme	Distributed	Message content, Geographical information	Dynamic Time Warping	Detection at regular intervals on a single vehicle.
IWOT-V [7]	Mixed	Historical interaction	Bayesian algorithm	Detection based on the vehicular trust value calculated by the central system based on indirect communication.
Xia et al. [34]	Distributed	Historical interaction	Weighted Markov Prediction	Detection at regular intervals on a single vehicle.

TABLE V: Simulation Calculation Time of Three Schemes

Scheme name	Simulation calculation time
Our Proposed Scheme	555.48s
IWOT-V [7]	106.84s
Xia et al. [34]	130.76s

TABLE VI: Performance Comparison with Our Previous Work

Parameters (POIs, Malicious, speed)	Our Proposed			Previous Work [13]		
	Detection Rate	Recall Rate	Precision Rate	Detection Rate	Recall Rate	Precision Rate
0.4, 40%, [15,25]	0.9633	0.9633	0.948	0.946	0.9172	0.9378
0.4, 50%, [15,25]	0.834	0.834	0.836	0.777	0.7238	0.7904
0.5, 50%, [15,25]	0.878	0.878	0.881	0.785	0.722	0.8068
0.6, 50%, [15,25]	0.863	0.863	0.866	0.79	0.721	0.819
0.7, 50%, [5,25]	0.882	0.887	0.88	0.7913	0.827	0.6913
0.7, 50%, [10,25]	0.882	0.886	0.881	0.7946	0.8343	0.7046

realistic time scale, the computing resources of single vehicle are sufficient to achieve the effect of real-time detection.

In terms of our simulation calculation time, since DTW needs to construct a distance matrix between each point in two long trajectory sequences during calculation, certain computing resources are required. Meantime, during the 6-h simulation time, every time the social relationship is updated based on the DTW value, the distance matrix between the earlier points in the trajectory sequence will be repeatedly calculated, resulting in a serious waste of computing resources. Since we focused on the detection rate, recall rate, and precision rate when implementing our scheme, there will be efforts to optimize the computation resources (e.g., to optimize data structure for trajectory processing) when implementing our scheme.

Our proposed scheme focuses on detecting attackers in VANETs. How to reroute vehicles after detecting traffic events is a research problem in civil engineering, while our focus is on detecting misbehaviors in VANETs from the scope of cyber security. Of course, the most important thing is to accurately identify whether the event (traffic jam, or accident) has occurred, so as to plan the vehicle route not passing through those points to alleviate the potential traffic congestion.

5) Performance Comparison with Our Previous Work

We implement our previous work [13] in this paper scenario, and compare the performance of our proposed scheme with that of previous work in Table VI. It can be seen that by adopting the improved geographic information calculation method and reputation system, the performance of our proposed scheme is improved under various experimental parameters compared with the performance of previous work. In addition, our previous work does not perform as well as previously expected, when dealing with newly added time-shift attacks and complex traffic scenario (i.e., multiple events). Our proposed scheme improves the detection rate, recall rate, and precision rate compared with previous work after adopting the improved method. Simulation experiments demonstrate the effectiveness of our proposed scheme.

VI. CONCLUSION

This paper explored an effective misbehavior detection scheme based on geographic information to guarantee the reliability of VANETs communication. It consisted of three algorithms, i.e., message trust measurement, social investiga-

tion construction, and reputation system. The message trust measurement calculates the trust value of message based on the content message. The social investigation construction focuses on the geographic information between vehicles. The reputation system dynamically adjusts the malicious value and introduces negative evaluation in the indirect social relationship. This paper constructs social relationships between vehicles through more effective calculation of geographic information. Meanwhile, this paper improves the applicability of proposed scheme in complex traffic based on traffic multi-event levels and reputation systems. Extensive simulations were conducted to evaluate the performance of proposed scheme. The simulation results showed that the proposed scheme could accurately identify normal and malicious vehicle with purposeful movements.

The future work can be investigated from two aspects. Firstly, the influence of vehicle social preference factors on geographic information could be improved. Secondly, the influence of vehicle identity on social trust could be considered.

VII. ACKNOWLEDGEMENT

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